

## Trainings ▾

## General Information

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**Note :** Due to the number of water inlet connection options, the following procedure has been commonized. The illustrations may not match the engine being serviced, but the procedures are the same.

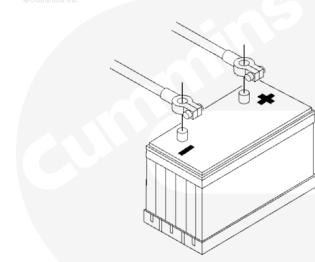
## Preparatory Steps

### ⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.

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### ⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

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### ⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

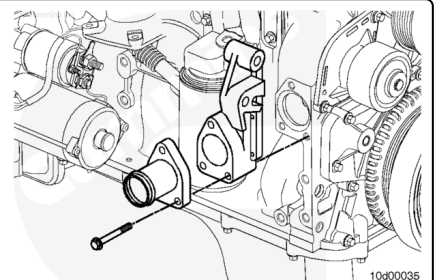
- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).

**Note : Some belt tensioners are mounted to the water inlet connection.**

- If required, remove the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- If required, remove the fan belt tensioner. Refer to Procedure 008-087 (</qs3/pubsys2/xml/en/procedures/40/40-008-087-tr.html>)
- Remove the lower radiator hose. Refer to Procedure 008-045 (</qs3/pubsys2/xml/en/procedures/40/40-008-045.html>)
- If required, disconnect any OEM coolant hoses. Refer to the OEM service manual
- If required, remove the alternator and alternator mounting brackets. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/40/40-013-001-tr.html>) and Procedure 013-003 (</qs3/pubsys2/xml/en/procedures/40/40-013-003-tr.html>).

## Remove

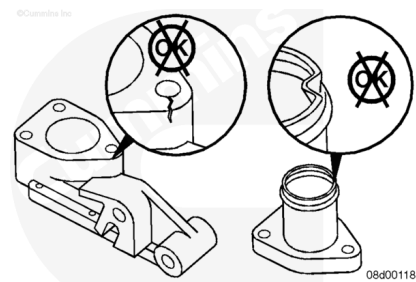
Remove the capscrews, water inlet connection, gasket, and rectangular sealing ring.



## Clean and Inspect for Reuse

Inspect the rectangular sealing ring for cracks.

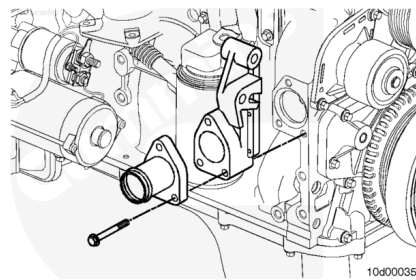
If evidence of leaking exists, replace the rectangular sealing ring.



## Install

Install the capscrews, water inlet connection, gasket, and rectangular sealing ring.

Align the roll pins against the front face of the cylinder block and tighten the capscrews.



|     |        |            |
|-----|--------|------------|
| M10 | 43 n.m | [32 ft-lb] |
| M12 | 80 n.m | [59 ft-lb] |

## Finishing Steps

### **⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

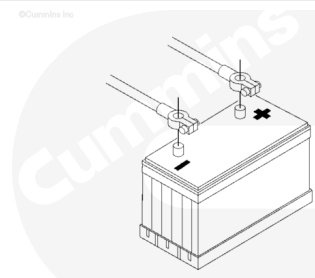


- If removed, install the fan belt tensioner. Refer to Procedure 008-087 (</qs3/pubsys2/xml/en/procedures/40/40-008-087-tr.html>)
- If removed, install the cooling fan drive belt. Refer to Procedure 008-002 (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- If removed, connect any OEM coolant hoses. Refer to OEM service manual
- Connect the lower radiator hose. Refer to Procedure 008-045 (</qs3/pubsys2/xml/en/procedures/40/40-008-045.html>)

- If removed, install the alternator and alternator mounting brackets. Refer to Procedure 013-001 (</qs3/pubsys2/xml/en/procedures/40/40-013-001-tr.html>) and Procedure 013-003 (</qs3/pubsys2/xml/en/procedures/40/40-013-003-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/40/40-008-018-tr.html>).

**⚠ WARNING ⚠**

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- Connect the batteries
- Operate the engine and check for leaks, alignment, and tension on the drive belt.